

Use Cases

1 Controlling RGB Light Panel

```
from pymycobot.mycobot320 import MyCobot320
import time

# The above needs to be written at the beginning of the code, which means
importing the project package

# MyCobot320 class initialization requires two parameters:
#   The first is the serial port string, such as:
#       linux: "/dev/ttyUSB0"
#       or "/dev/ttyACM0"
#       windows: "COM3"
#   The second is the baud rate:
#       M5 version is: 115200
#
#   Example:
#       mycobot-M5:
#       linux:
#           mc = MyCobot320("/dev/ttyUSB0", 115200)
#       or mc = MyCobot320("/dev/ttyACM0", 115200)
#       windows:
#           mc = MyCobot320("COM3", 115200)

# Initialize a MyCobot320 object
# Create object code here for windows version
mc = MyCobot320("COM13", 115200)

i = 7
# loop 7 times
while i > 0:
    mc.set_color(0, 0, 255) # blue light on
    time.sleep(2) # wait for 2 seconds
    mc.set_color(255, 0, 0) # red light on
    time.sleep(2) # wait for 2 seconds
    mc.set_color(0, 255, 0) # green light on
    time.sleep(2) # wait for 2 seconds
    i -= 1
```

2 Controlling Arms to Move Them to Starting Point

```
from pymycobot.mycobot320 import MyCobot320

mc = MyCobot320("COM3", 115200)

# Check whether the program can be burned into the robot arm
if mc.is_controller_connected() != 1:
    print("Please connect the robot arm correctly for program writing")
    exit(0)

# Fine-tune the robotic arm to ensure that all the bayonets are aligned in the
adjusted position
# Subject to the alignment of the mechanical arm bayonet, this is only a case
mc.send_angles([0, 0, 0, 0, 0, 0], 30)
```

3 Single-Joint Motion

```
from pymycobot.mycobot320 import MyCobot320
import time

mc = MyCobot320('COM3', 115200)

# Robotic arm recovery
mc.send_angles([0, 0, 0, 0, 0, 0], 40)
time.sleep(3)

# Control joint 3 to move 70°
mc.send_angle(3, 70, 40)
time.sleep(3)

# Control joint 4 movement -70°
mc.send_angle(4, -70, 40)
time.sleep(3)

# Control joint 1 to move 90°
mc.send_angle(1, 90, 40)
time.sleep(3)

# Control joint 5 movement -90°
mc.send_angle(5, -90, 40)
time.sleep(3)

# Control joint 5 to move 90°
mc.send_angle(5, 90, 40)
time.sleep(3)
```

4 Multi-Joint Motion

```
import time
from pymycobot import MyCobot320

# MyCobot320 class initialization requires two parameters:
```

```

# The first is the serial port string, such as:
#     linux: "/dev/ttyUSB0"
#         or "/dev/ttyACM0"
#     windows: "COM3"
# The second is the baud rate:115200
#
# Example:
#     mycobot-M5:
#         linux:
#             mc = MyCobot320("/dev/ttyUSB0", 115200)
#         windows:
#             mc = MyCobot320("COM3", 115200)
#
# Initialize a MyCobot320 object
mc = MyCobot320('COM3', 115200)

# Robotic arm recovery
mc.send_angles([0, 0, 0, 0, 0, 0], 50)
time.sleep(2.5)

# Control different angles of rotation of multiple joints
mc.send_angles([90, 45, -90, 90, -90, 90], 50)
time.sleep(2.5)

# Return the robotic arm to zero
mc.send_angles([0, 0, 0, 0, 0, 0], 50)
time.sleep(2.5)

# Control different angles of rotation of multiple joints
mc.send_angles([-90, -45, 90, -90, 90, -90], 50)
time.sleep(2.5)

```

5 Coordinate control

```

from pymycobot.mycobot320 import MyCobot320
import time

# MyCobot320 class initialization requires two parameters:
# The first one is the serial port string, such as:
#     # linux: "/dev/ttyUSB0"
#     # windows: "COM3"
# The second is the baud rate:
# M5 version is: 115200
#     # like:
#     # mycobot-M5:
#         # linux:
#         # mc = MyCobot320("/dev/ttyUSB0", 115200)
#         # windows:
#         # mc = MyCobot320("COM3", 115200)

# Initialize a MyCobot320 object
# Create the object code below for the windows version
mc = MyCobot320("COM3", 115200)

```

```

# Get the coordinates and posture of the current head
coords = mc.get_coords()
print(coords)

# Send angular motion to a certain attitude for coordinate control, with a speed
of 50
mc.send_angles([0, 0, -90, 0, 90, 0], 50)
time.sleep(2.5)

# Intelligent planning of the route, allowing the head to reach the coordinates
of [-2.3, -153.2, 400.8] in a linear manner, and maintain the posture of [-120.0,
0.7, 179.73], with a speed of 40mm/s
mc.send_coords([-2.3, -153.2, 400.8, -120.0, 0.7, 179.73], 40, 1)

# Set the waiting time to 1.5 seconds
time.sleep(1.5)

# Intelligent planning of the route, allowing the head to reach the coordinates
of [0.0, -120.4, 500.3] in a linear manner, and maintain the posture of [-70.81,
-22.17, -163.49], with a speed of 50mm/s
mc.send_coords([0.0, -120.4, 500.3, -70.81, -22.17, -163.49], 50, 1)

# Set the waiting time to 1.5 seconds
time.sleep(1.5)

# Only change the x-coordinate of the head, setting the x-coordinate of the head
to 100. Let it intelligently plan the route to move the head to the changed
position at a speed of 70mm/s
mc.send_coord(Coord.X.value, 100, 70)

```

6 Control the robot arm to swing left and right

```

from pymycobot.mycobot320 import MyCobot320

import time
# Initialize a MyCobot320 object

# M5 version
mc = MyCobot320("COM3", 115200)
# Get the coordinates of the current position
angle_datas = mc.get_angles()
print(angle_datas)

# Use a sequence to pass the coordinate parameters and let the robot move to the
specified position
mc.send_angles([0, 0, 0, 0, 0, 0], 50)
print(mc.is_paused())
# Set the waiting time to ensure that the robot has reached the specified
position
# while not mc.is_paused():
time.sleep(2.5)

# Move joint 1 to the position of 90
mc.send_angle(1, 90, 50)

```

```

# Set the waiting time to ensure that the robot has reached the specified
position
time.sleep(2)

# Set the number of loops
num = 5

# Let the robot swing left and right
while num > 0:
    # Move joint 2 to position 50
    mc.send_angle(2, 50, 50)
    # Set the waiting time to ensure that the robot has reached the specified
    position
    time.sleep(1.5)
    # Move joint 2 to position -50
    mc.send_angle(2, -50, 50)
    # Set the waiting time to ensure that the robot has reached the specified
    position
    time.sleep(1.5)
    num -= 1

# Retract the robot. You can manually swing the robot, and then use the
get_angles() function to get the coordinate sequence,
# Use this function to make the robot reach the position you want.
mc.send_angles([88.68, -135, 145, -128.05, -9.93, -15.29], 50)

# Set the waiting time to ensure that the robot arm has reached the specified
position
time.sleep(2.5)
# Let the robot arm relax and you can swing it manually
mc.release_all_servos()

```

7 Controlling the robotic arm to dance

```

from pymycobot.mycobot320 import MyCobot320
import time

if __name__ == '__main__':
    # MyCobot320 class initialization requires two parameters:
    # The first is the serial port string, such as:
    #   Linux: "/dev/ttyUSB0"
    #   Windows: "COM3"
    # The second is the baud rate:
    #   M5 version: 115200

    # Example:
    # mycobot-M5:
    #   Linux:
    #     mc = MyCobot320("/dev/ttyUSB0", 115200)
    #   Windows:
    #     mc = MyCobot320("COM3", 115200)

    # Initialize a MyCobot320 object
    # M5 version
    mc = MyCobot320("COM3", 115200)

```

```

# Set the start time
start = time.time()
# Let the robot reach the specified position
mc.send_angles([-1.49, 115, -145, 30, -33.42, 137.9], 80)
# Determine whether it has reached the specified position
while not mc.is_in_position([-1.49, 115, -145, 30, -33.42, 137.9], 0):
    # Let the robot resume movement
    mc.resume()
    # Let the robot move for 0.5s
    time.sleep(0.5)
    # Pause the movement of the robot
    mc.pause()
    # Determine whether the movement has timed out
    if time.time() - start > 3:
        break
# Set the start time
start = time.time()
# Let the exercise last for 30 seconds
while time.time() - start < 30:
    # Let the robot quickly reach this position
    mc.send_angles([-1.49, 115, -145, 30, -33.42, 137.9], 80)
    # Set the color of the light to [0,0,50]
    mc.set_color(0, 0, 50)
    time.sleep(0.7)
    # Let the robot quickly reach this position
    mc.send_angles([-1.49, 55, -145, 80, 33.42, 137.9], 80)
    # Set the color of the light to [0,50,0]
    mc.set_color(0, 50, 0)
    time.sleep(0.7)

```

8 Controlling Gripper

```

from pymycobot.mycobot320 import MyCobot320
import time

# MyCobot320 class initialization requires two parameters:
#   The first is the serial port string, such as:
#       linux: "/dev/ttyUSB0"
#       or "/dev/ttyACM0"
#       windows: "COM3"
#   The second is the baud rate::
#       M5 version is: 115200
#
#   such as:
#       mycobot-M5:
#           linux:
#               mc = MyCobot320("/dev/ttyUSB0", 115200)
#               or mc = MyCobot320("/dev/ttyACM0", 115200)
#           windows:
#               mc = MyCobot320("COM3", 115200)

# Initialize a MyCobot320 object
mc = MyCobot320("COM13", 115200)
# make it move to zero position
mc.send_angles([0, 0, 0, 0, 0, 0], 40)

```

```

time.sleep(3)
# Pro adaptive gripper needs to be set to transparent transmission mode before
use
mc.set_gripper_mode(0)
time.sleep(1.5)

num = 5
while num > 0:
    # Set the state of the gripper to quickly open the gripper at a speed of 70,
    Parameter 1 represents pro adaptive gripper type
    mc.set_gripper_state(0, 70, 1)
    time.sleep(3)
    # Set the state of the gripper to quickly close the gripper at a speed of 70,
    Parameter 1 represents pro adaptive gripper type
    mc.set_gripper_state(1, 70, 1)
    time.sleep(3)
    num -= 1

num = 3
while num > 0:
    # Set the gripper value to 0 and the speed to 70, Parameter 1 represents pro
    adaptive gripper type
    mc.set_gripper_value(0, 70, 1)
    time.sleep(3)
    # Set the gripper value to 50 and the speed to 70, Parameter 1 represents pro
    adaptive gripper type
    mc.set_gripper_value(50, 70, 1)
    time.sleep(3)
    # Set the gripper value to 0 and the speed to 70, Parameter 1 represents pro
    adaptive gripper type
    mc.set_gripper_value(0, 70, 1)
    time.sleep(3)
    num -= 1

# Get the value of the gripper
print("gripper value:", mc.get_gripper_value())

```